Monte Carlo
The Spectral Dimension of a Triangulation

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Abstract

In this research we try to numerically estimate the spectral dimension D_{spec} for two-dimensional quantum gravity, which has been demonstrated to be equal to 2. Utilizing triangulations generated by four different models, we use two different techniques to gather data and then derive the spectral dimension. A classic random walk and the adjacency matrix.

We find that the errors calculated are largely underestimated, but the choice of interval in which to evaluate D_{spec} plays a major role, due to discreteness and finite size effects.

Our estimations range between $D_{spec} = 1.79$ and $D_{spec} = 1.91$, depending on model and method, but our errors are estimated to be at most $\sigma = 0.03$.

1 Introduction

One of the main objectives of studying two-dimensional quantum gravity is to gain an understanding of the fractal structure of space-time. In particular, the two-dimensional case (rather than more general four-dimensional spaces) allows both analytical and numerical tools so it is an interesting laboratory.

A common way to study a d-dimensional manifold is to construct a triangulation by gluing together N d-simplices so that the final structure has the same topology as the original manifold. We can then intuitively use triangulations as a discretized version of the two-dimensional manifold we would like to study, given that we take a sufficiently large number N of triangles. To be more specific: the triangulations we use are closed (without boundaries) planar maps, without any ”handles” that would make them torus-like, so topologically equivalent to 3-d spheres. In this paper we look at 4 different models[1] for generating these graphs: Uniform[2], Spanning-tree decorated[34], Bipolar-oriented[5], Schnyder-Wood-decorated[6].

Our main focus is the *spectral dimension*. In a somewhat similar way to the Hausdorff dimension that we studied in the lectures, the spectral dimension is an indicator of geometry and topology of space-time and allows to compare the description of quantum geometry in various approaches to quantum gravity. This is possible because it can be defined not only on smooth geometries but also on discrete ones[7]. The spectral dimension is not dependent on the specific triangulation: for all four models the spectral dimension is analytically proven to be 2[1].

The spectral dimension can also be numerically calculated via its relation with the diffusion of matter on the surface, which, in turn, may be simulated by a random walk over the surface. The random walk starts at a random triangle (position $x = 0$) at time $t = 0$. After a time t the random walk has a probability of $\Psi(t;x)$ to be at position x. We should then calculate the probability that the random walk is at its starting point after time t: $\Psi(t;0)$. The average of the return probabilities for (all) different triangles, $\overline{\Psi}(t;0)$, is related to the spectral dimension[8]

$$D_{spec} = -2 \lim_{t\to\infty} \frac{\log \overline{\Psi}(t;0)}{\log t} \tag{1}$$

The time t is just the number of steps taken by the random walk. This method to calculate D_{spec} looks at the return probability if the random walk length t goes to infinity, but the longer the random walk the longer it takes to calculate. Therefore it is easier to use an other definition.

In the continuum limit of diffusion 1 can be rewritten as (see[8])

$$\overline{\Psi}(t;0) = k \cdot t^{-D_{spec}/2} \tag{2}$$

Here t must not be too small or our approximation of the continuum space with a discrete lattice would not hold. But if t is too large compared to the size of the triangulation the return probability will be determined by the number of triangles and eventually converges to a uniform distribution. For a small enough t the effect of the finite size can be neglected and 2 can be used. This means the random walks can be shorter than required from 1.

In practice we choose two different methods to calculate the return probability of a random walk: actually running random walks over the surface; and calculating the adjacency matrix of the triangulation.

The first method runs a lot of random walks on the surfaces of different triangulation and sees how often it returns to its starting point for every time t. From this, equation 2 can be fitted to get the spectral dimension.

For the second method the adjacency matrix is used. It is a square matrix with the size given by the total number of triangles in the triangulation, in which the rows and columns represent each node of the graph. All non-zero entries then represent a link (i.e. a shared edge) between two elements.

As we will see into more details in Adjacency Matrix, multiplying this matrix by itself t times, and taking the diagonal elements gives us the return probability after t steps.

The next section describes how the triangles are constructed and how both methods are implemented. Section Error estimation shows the errors that we calculate and section Results the results of both methods and their comparison.

2 Methods

This describes how the triangles are generated and how the two methods used to calculate the spectral dimension are implemented. For both methods the spectral dimension is calculated for every triangulation model separately.

There are two main options on the way to approach the random walk: either having the path be from face to face, through the edges, or from vertex to vertex, along the edges. This applies to both methods. We decided to implement the first scheme, since the adjacency of triangles is given by the triangle generator, see below. Calculating the adjacency of the vertices in this way makes the program for the random walk really slow. A second factor is that every triangle face always has exactly three links (unlike vertices), which instead improves the computational efficiency.

2.1 Construction of Triangulations

To produce the triangulations we use the black box generator from exercise sheet 8 of the Monte Carlo course, using one of the four models (U, W, S, B) mentioned in Introduction.

Computationally, triangulations are stored as an array of integers that represents the edges and their connections, e.g. if i and j are connected, the value of the i^{th} element is j. Triangles are also numbered and the edges follow the order of the triangles so that the index divided by 3 (`index//3`) is the triangle number.

We found that this generator is quite fast even for large triangulations of size 2^{14} $(16,384)$ or 2^{15} $(32,768)$. The generation of large triangulations is therefore not a limitation to the calculation time.

For comparison we found that creating an independent triangulation with a Markov Chain is much slower (~ 3000 times slower) and hence not used in this paper.

Although producing these maps is not computationally expensive, the actual random walks and the matrix multiplication (in particular this last one) are very costly; after first tests we determined that 2^{14} is a sufficiently large size. (More in the next sections Random Walk and Adjacency Matrix)

2.2 Random Walk

For every random walk a new triangulation is generated and a starting point is randomly (uniformly) chosen. Every next step is then easily chosen from the neighbouring trianglesuntil the maximum path length, t_{max}, is reached. At every step we check to see if it coincides with the starting point. Repeating this process and averaging gives us an array that approximates the return probability $\overline{\Psi}(t;0)$, for every t. Finally we iterate for the four different models of generating triangulations.

Because every random walk is conducted on a new triangulation with a new starting point, random walks are completely independent from each other. On the other hand, if a random walk at time t lands in a point very "far away" from its starting triangle, the chance of coming back to the origin at $t+1$ is likely zero. The opposite is also true: being close to the initial triangle at time t makes the return probability at immediately subsequent ts more likely. It can clearly be seen how the sequence of steps may be

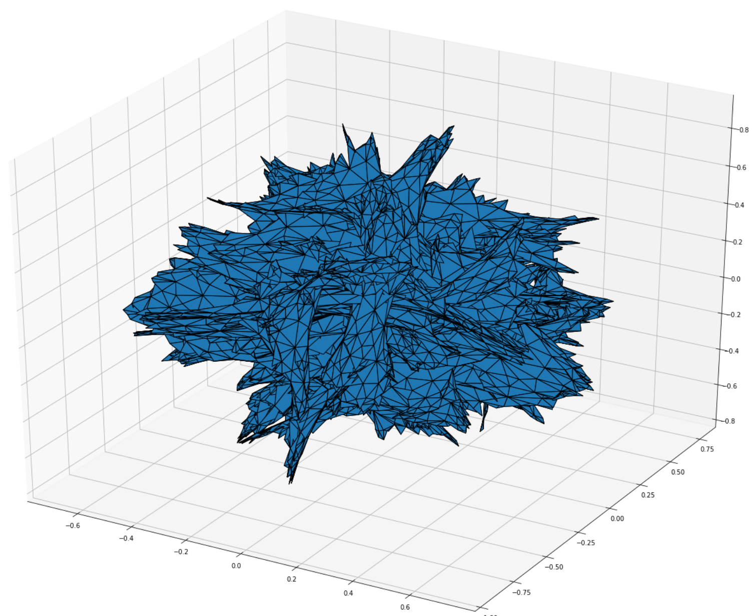


Figure 1: An example of a Schnyder-Wood-decorated triangulation of the same size (16384) as the actual simulations

auto-correlated, so, to account for this, we make sure to have a sufficient number of independent runs.

Another issue to watch out for is the finite size of the triangulations: if the maximum length path t_{max} becomes close to or bigger than said size, $\overline{\Psi}(t;0)$ approaches a uniform distribution for all t. Again, after the first trials, the final simulations are run for $400,000$ iterations, with $t_{max} = 10,000$ and a triangulation with size $N = 2^{14} = 16384$.

The average return probabilities are plotted against t and fitted using 2 and `scipy.optimize.curve_fit` to get the value of the spectral dimension. The maximum t is small enough compared to the triangle size to neglect the finite size effect. The data starts from $t = 0$, since these steps has to be done for the random walk anyway and it is inexpensive to record. The fit does not start at $t = 0$ to ignore the effects of discretization at small scale. Which t to start from does influence the resulting fit and thus the measured spectral dimension. The same is true for the matrix method.

2.3 Adjacency Matrix

The adjacency matrix is used to represent a finite graph: the elements of the matrix indicate whether pairs of triangles are adjacent or not.

By assigning these entries to $1/3$ (uniform probability for every random walk step) and then multiplying t times this matrix with itself we get a resulting matrix that represents the probability of every starting point i to be found at position j after t steps $\mathbf{M}(t) := \mathbf{M}^t$.

The average of the $Trace$ (so simply the average diagonal element) is related to the return probability via.

$\langle \mathrm{Tr}\, \mathbf{M}(t) \rangle = \Psi(t;0)$The spectral dimension can therefore be calculated with a very similar function.[9]

$$\langle \mathrm{Tr}\, \mathbf{M}(t) \rangle \sim t^{-D_{spec}/2} \tag{3}$$

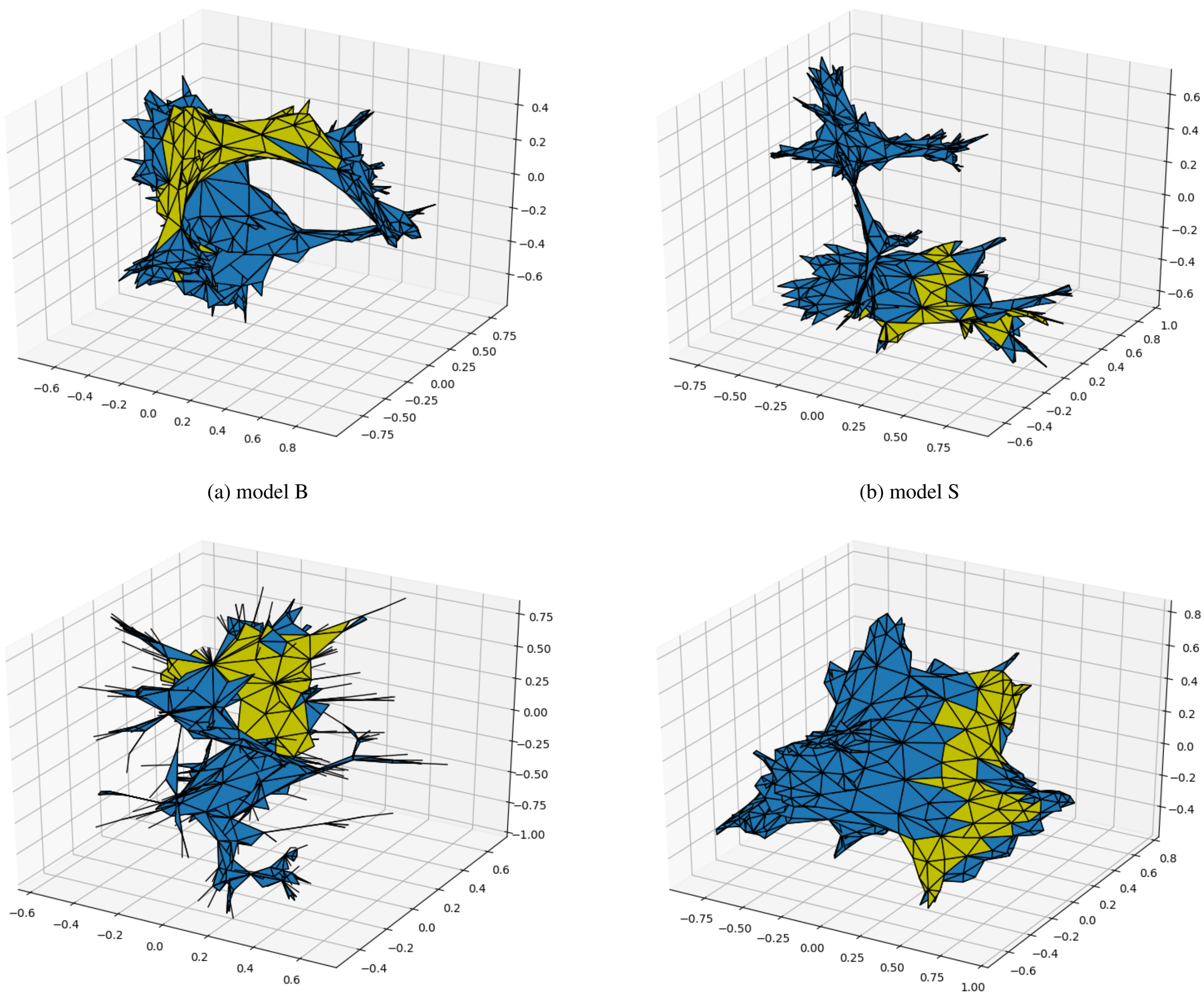
This method has some advantages:

- it takes into consideration the whole triangulation, so each single result is exact
- it only requires a single graph per measurement

but unfortunately it also has some serious downsides, mainly due to computational limitations:

- The matrix applies to only one triangulation. To get a good estimate, it still needs multiple runs.
- The matrix size itself ($n = 2^{14} = 16384$) can be an issue from the storage point of view, since it contains $2.7 \cdot 10^8$ `float` values
- Matrix multiplication is particularly demanding, as it scales roughly as n^3

One way we use to reduce this drawbacks is to adopt the COO (coordinates) representation of matrices, which represents it as an array of triplets `(i,j,v)`, i.e. (row,



(a) model B

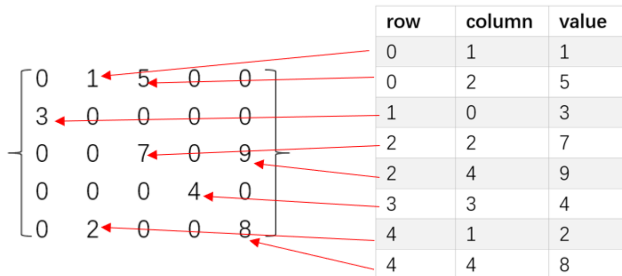
(b) model S

(c) model U

(d) model W

Figure 2: Examples of triangulations built using the four different models, and a random walk over the surface (every triangle in the path is painted yellow). The size is smaller than in our simulations: $N = 2^{10} = 1024$.

column, value). Since the matrix is mostly sparse (at construction every row and column has only three entries), this allows for much more efficient calculation.



row	column	value
0	1	1
0	2	5
1	0	3
2	2	7
2	4	9
3	3	4
4	1	2
4	4	8

Figure 3: Example of how the *COO notation* works

In the end, even though we are only able to reach $t = 1000$, for just 10 reruns each, the resulting data proves to be fairly consistent and a valuable comparison with the classical random walk method.

3 Error estimation

We speculate that the error of the spectral dimension depends on two things: the error in the measured return probabilities and the variation in the D_{spec} as an effect of changing the starting point of the fit. The way to calculate the error of the return probability is different for the the adjacency matrix and the random walk method. The way to calculate the errors as effect of the starting point for the fit is the same for both.

3.1 Error of the return probability

For the matrix method the return probabilities of a single triangulation are exact and completely independent between different t. The triangulations are also completely independent of each other. The error of the average return probability is therefore just the standard deviation of the return probability. For every t we have a (potentially) different value of standard deviation.

For the random walk, the measurements of the return probabilities are not completely independent. The different random walks are independent, but the return probabilities of neighbouring t values are not. By running a big number of random walks we do reduce this effect. If the values were completely independent, the error would be simply a binomial error.

We also use the *bootstrapping* method mentioned in the lectures, getting very similar errors. This method resamples from the values measured (it can chose a value more than once), creating new sets of data with the same distributions. In our simulation we settle for t between $100-1000$ and 20 new resampled datasets, but, even increasing this quantity by a wide margin, the resulting σ is unlikely to change drastically. In practice, the original samples are in the form of arrays of 1s and 0s (if the random walk returns or does not) for each t value. Which means that in our case it is just a binomial distribution and, as such, the error is simply:

$$\sigma_{bin} = \sqrt{\frac{pq}{n}} \tag{4}$$

Where p is the probability of the event, $q = 1 - p$, and n the number of trials. These errors are used by `scipy.optimize.curve_fit` to calculate the error of the spectral dimension from the fit.

But, as we will see in Results, our errors seem to be considerably underestimated and none of the previous methods improve our evaluation.

3.2 Error as effect of the starting point of the fit

A notable remark is that we find that the choice of the interval ($start - end$) in which to fit the data has a big impact on the result. For the endpoint the condition is, as mentioned earlier, that the effect of the finite size of the triangulation must be negligible. This is still true for the maximum t, since we chose t_{max} to be reasonably smaller than N_{triang}. On the other hand, the start value should not be too small, due to discrete effects, but there is not one exact value to set it. Varying this quantity influences the value of the spectral dimension by a substantial amount. Therefore we fit the function for different starting points and look at the variation of the spectral dimension, see graph 4a.

At much higher starting values, D_{spec} becomes almost chaotic as a result of the small amount of data and the inconvenient interval in which the `curve_fit` has to operate. This could have several causes. It could be a result of the fact that there are much less data points between 7000 and 10000 than if you start at a lower t. This seems very unlikely since there are still a lot of datapoints. The difference between them, however, becomes a lot smaller. This could make it more difficult to fit the function starting at higher t. It could also be an effect of the graph flattening out. This would be an indication that the finite size of the triangulation effects the return probability of the longer random walks. This is however not visible in the graphs shown in section 4.

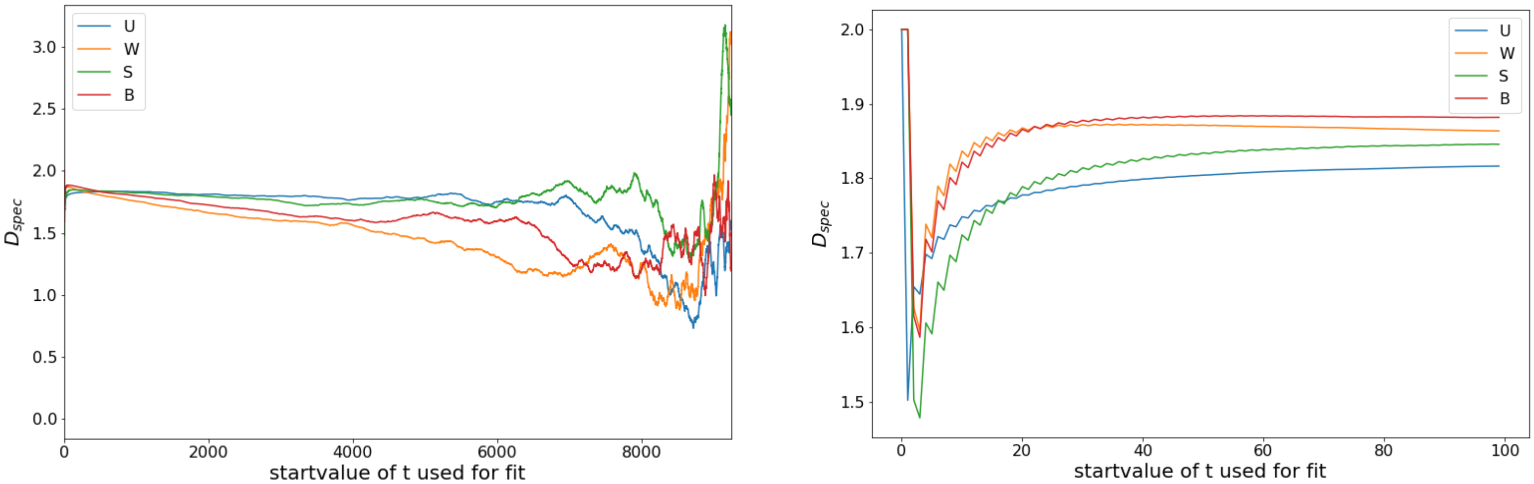
Graph 4b shows the variations in D_{spec} if the fits starts at random walk lengths t that are to small too approach continuum diffusion. These low values of t are skipped.

To take this range of D_{spec} into account the fit is done with different starting points. The starting value of t should be approximately higher than 100, see figure 4a, and lower than 1000, see figure 4b, to get a reasonable fit. We chose the upper limit for the staring value to be $t = 1000$ to keep as much data points as possible. Fitting the data gives different values of the spectral dimension for the different starting points. From these, the standard deviation can be calculated for the four models. For the random walk this is:

$\sigma_U = 0.549 \quad \sigma_W = 0.541 \quad \sigma_S = 0.551 \quad \sigma_B = 0.551.$

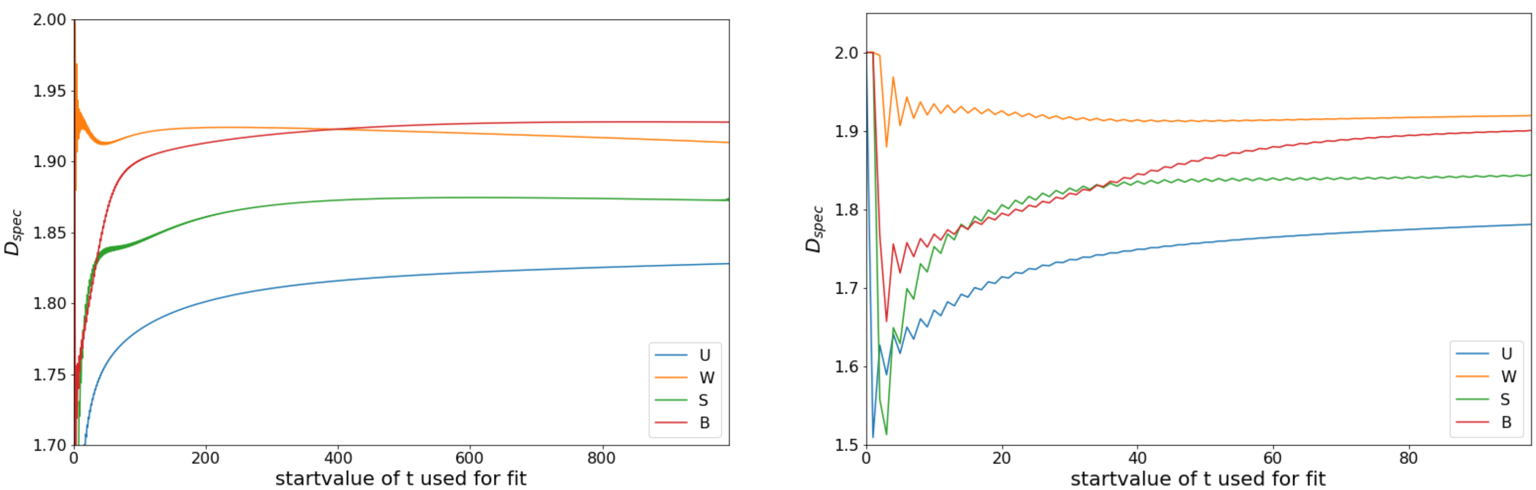
For the matrix method this is:

$\sigma_U = 0.0056 \quad \sigma_W = 0.0013 \quad \sigma_S = 0.0037 \quad \sigma_B = 0.0039.$



(a) (b)

Figure 4: The varying spectral dimension, D_{spec}, from the random walk for different t values to start the fit from. All fits are until $t = 10000$, only the starting point is varied. 4a is the full range of starting points and 4b is zoomed in on the first 100 t. In 4b the effect of the discretization is clearly visible.



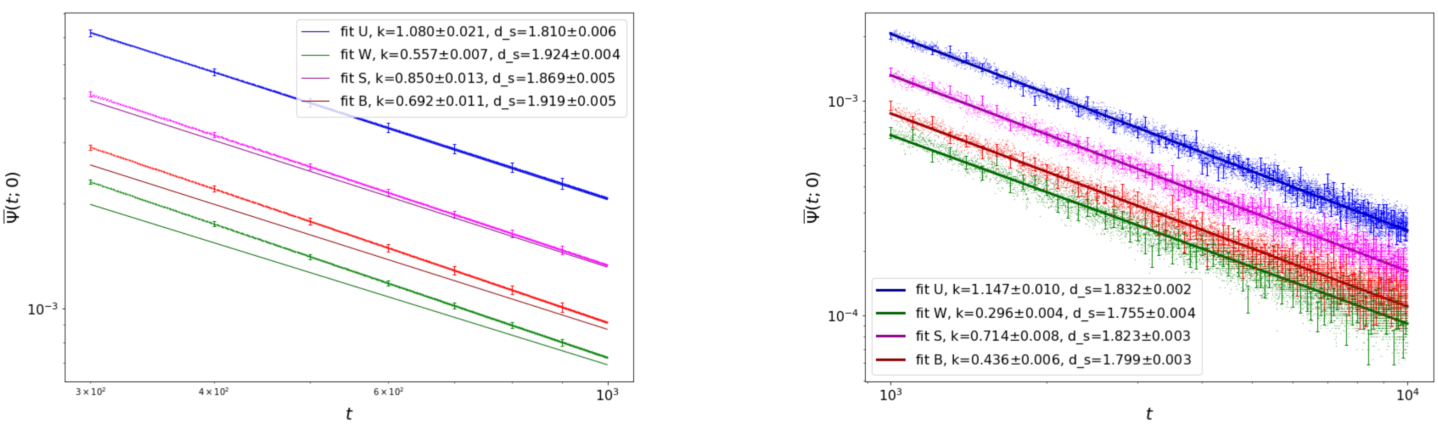
(a) (b)

Figure 5: Again the varying spectral dimensions, D_{spec}, but for the matrix method. All fits are till $t = 1000$. Left is again the full range and right zoomed in for t till 100. Here the effects of the of the discretization are again clearly visible.

4 Results

The random walk method needs much less computing time than the matrix method. For this reason the matrix method only has return probabilities for t between 0 and 1000. For the random walk we are able to get to maximum length $t = 10,000$. The resulting return probabilities for the matrix method and random walk method are shown in 6a and 6b respectively. The function $\overline{\Psi}(t;0) = k \cdot t^{-D_{spec}/2}$ (equation 2 from Introduction) is fitted, taking into account the errors of the matrix and random walk data. The values of k and D_{spec}, and their relative errors, are given in the legends.

These value of the spectral dimension are below there analytically proven value of $D_{spec} = 2$. The errors are also very small, so $D_{spec} = 2$ does not fall in the error region of the measured values.



(a) Adjacency matrix

(b) Random walk

Figure 6: Here is the resulting data: 6a is obtained from the adjacency matrix strategy and shows the data for the four models and their respective fit between $(300, 1000)$. It is quite evident that some of the fit are off by a meaningful amount: this induces us to think that the lower limit for t may be underestimated. 6b shows the results from the random walk method between $(10^3, 10^4)$. For clarity purposes the error bars are only shown every 100 steps

For the matrix method the fit seems to work well with the 'U' (Uniform) and 'W' (Schnyder-Wood-decorated) models, but not with the 'S' (Spanning-tree decorated) and 'B' (Bipolar-oriented) models. This seems to indicate that a random walk of 1000 steps is too short to calculate the spectral dimension. The length t between 300 and 1000 is not large enough to approximate a continuum diffusion for equation 2.

For the random walk method the fits seems to work fine, but, as we saw, the values of the D_{spec} are not sufficiently close to 2. The random walks should be long enough, but it could be that it becomes too comparable with the size of the triangulation. To see when this happens we plotted a random walk of length 1000 on different sized triangulations. These are shown in figure 7. The fits for the random walk on triangulation sizes 500 and 1000 are clearly different from the ones on triangulation sizes 2000 and larger. The fit of the random walk on a triangulation of 1600 is very close to the fits of larger triangulations, but there is a difference.

The random walk of $10,000$ from above is on a triangulation of size $16,000$ so this is the same ratio as a random walk of 1000 on a surface of 1600 triangles. It could therefore be that the finite size effect is not negligible in the return probability at t values close to $t = 10,000$. This would influence the D_{spec} value.

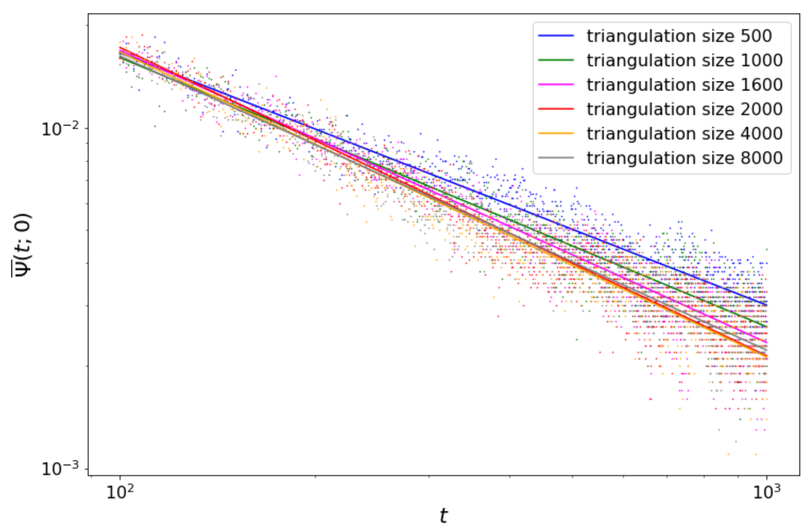


Figure 7: This shows how the fit of the function is influenced by the triangulation size. The return probabilities is calculated for a random walk of a constant length on different sizes of triangulations. The $\overline{\Psi}(t;0)$ is calculated with the random walk method with maximum length $t = 1000$ and 10000 runs. The fit is from $t = 100$ till $t = 1000$. There is a distinguishable difference between the size ≥ 2000 and the size 1000 and 500. This is the effect of the finite size of the triangulation. The fit of random walk on the 1600 size triangulation (same relation path/size as our simulation) seems to be not too impacted. The small dots are the return probabilities measured, the color matches the fits.

The return probabilities gathered with the random walk method and the adjacency matrix are very similar in the range t is $100 - 1000$, as in 8. This indicates that the random walk method has a sufficient number of trials and the matrix method is precise enough with only 10 triangulations.

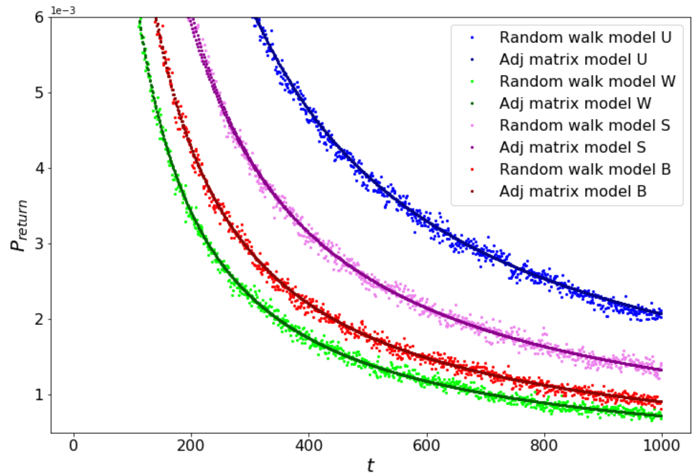


Figure 8: This shows both the random walk and the adjacency matrix data between $(0, 1000)$. As we expect, the two different methods have almost identical curves

Above only one fit is done to calculate the D_{spec}. There is, however, no clear point best for the fit to start. Therefor fits are done in a range of starting values for t $(100-1000)$. Averaging over these results gives D_{spec} from multiple fits. The error here is via the second method described in the section errors. This is stil to far from 2.

Models	Random Walk t_{start} in $(100, 1000)$	Adjacency matrix t_{start} in $(100, 300)$
U	$D_{spec} = 1.8309 \pm 0.0040$ $k = 1.0429 \pm 0.0181$	$D_{spec} = 1.7994 \pm 0.0081$ $k = 1.0429 \pm 0.0267$
W	$D_{spec} = 1.8021 \pm 0.0294$ $k = 0.5562 \pm 0.0444$	$D_{spec} = 1.9231 \pm 0.0011$ $k = 0.5562 \pm 0.0019$
S	$D_{spec} = 1.8362 \pm 0.0073$ $k = 0.8233 \pm 0.0227$	$D_{spec} = 1.8591 \pm 0.0075$ $k = 0.8233 \pm 0.0019$
B	$D_{spec} = 1.8335 \pm 0.0242$ $k = 0.6768 \pm 0.0511$	$D_{spec} = 1.9120 \pm 0.0049$ $k = 0.6768 \pm 0.0104$

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